# IY3660 – Applications of Cryptography

## Week 3 Workshop

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# Activity 1: Mini-essays

- In the summative assessments for this module I might ask you to write a mini-essay on a specific topic.

- General advice
  1. Recall the definition/contextualise the topic
  2. Answer precisely (compare, justify, argue..)
  3. Stay within the word limit
  4. Use your own words and ideas

# Examples – write 250-word essay

- Stream Ciphers vs. Block Ciphers: Which is More Appropriate for Different Applications?

- Cryptographic Strength and Key Length: How Strong is "Strong Enough”

# Activity 2: Example: “matrix” encryption

Imagine a scheme that encrypts $n$-bit strings $P$ by considering them as a $n\ x\ 1$ column vectors, and multiplying by an invertible secret $n\ x\ n$ matrix $M$

$$C = M . P$$

**Attack strategy (known-plaintext attack) to recover the secret matrix $\boldsymbol{M}$:**

- Among the known plaintexts, select any $n$ linear independent vectors $P_i$ .
- Construct a system of $n^2$ linear equations from the corresponding $n$ ciphertexts $C_i$
- Solve the system to recover $M$

*WHAT IS THE ATTACK COMPLEXITY?*

*Namely, how much data is used, how much memory and how much time is needed?*

# Activity 3: Quick revision

- Formalise the encryption and decryption steps of one roundo of Feistel cipher.